Independent University, Bangladesh

SETS- Computer Science and Engineering

Course Name:	Introduction to Robotics	**Course Code:**	CSE 426
Semester:	Summer 2026	**Section:**	1
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Assignment No:	1

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Submission Link (MS forms):			
Submission Date:	16 June 2026	**Due Date:**	16 June 2026

Question: Complete the following table after going through the datasheet/ online resources of the following micro-controllers:

	Arduino UNO	ATMega328	PIC24	PIC33	STM32	ESP32	Raspberry Pi 5
Architecture type	8 bit AVR	8 bit AVR RISC	16 bit Modified Harvard	16 bit Digital Signal Controller	32-bit ARM Cortex-M3	32-bit Xtensa LX6 Dual Core	64 bit ARM Cortex-A76 SBC
Number of pins	14 Digital 6 Analog	28 pins	28 pins	28pins	48 pins	30/38 pins	40 GPIO Pins
Processing Speed (MIPS)	16MHz	16MHz/upto 20 MHz	16 MIPS	70 MIPS	72MHz	up to 240MHz	2.4 GHz Quad Core
Program flash memory (bytes)	32KB	32KB	64 KB	256KB	64 KB	usually 4MB module flash	MicroSD Storage
Operating voltage range (V)	5v	1.8-5.5v	2.0V-3.6V	3.0-3.6v	2.0-3.6V	3.0-3.6	5v input
No. of PWM channels	6	6	5 output Compare PWM	Up to 10 PWM outputs	12 PWM, capable timer channels	up to 16 LEDC PWM channels	Software PWM via GPIO
Communication Interfaces	UART, SPI I2C, USB	URAT, SPI I2C	URAT, SPI I2C, PMP	URAT, SPI I2C, CAN	USART, SPI I2C, USB, CAN	UART, SPI Wifi Bluetooth	USB, Ethernet, Wifi. Bluetooth GPIO, URAT, I2C
Original Cost / Price in BD (with reference link)*	900-1500	250-400	500-900	700-1200	300-600	450-900	8500-12000

<u>Ref:</u> Robotics BD. Raspberry Pi 5 is Depending on the RAM so, based on that Prices can be Different.

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1. What is Microprocessor and Microcontroller? Write down the differences in detail with examples? (Hand Written)

2. Draw Arduino UNO, ESP 32, Arduino Mega 328 and Raspberry Pi 5 PIN Diagram. (Free hand drawing)

3. Which Microcontroller board are you going to use in your course project? Explain with technical reasons. (Hand Written)

Ans to the Q! NO: 1

Microprocessor: is a integrated circuit that mainly contains the CPU. It performs arithmetic, logic and control operations, but it needs external memory. input/output ports and other peripherals to work as a complete system. Microprocessor are mainly used in Computers laptops and high-performance systems.

example: Intel core i5, i7, AMD Ryzen, ARM Cortex-A processors.

Microcontroller: is a compact computer on a single chip. It Contains CPU, RAM, flash memory, timers input/output pins and communication modules inside one IC. It is mainly used for controlling specific tasks in embedded systems.

example: ATMega 328P, ESP32, STM32F103, PIC24, dsPIC 33.

P.T.O

Difference between Microprocessor and Microcontroller :-

Topic	Microprocessor	Microcontroller
main function	General-purpose processing	Specific Control task
Components	CPU only	CPU, Memory and peripherals in one chip
Memory	external RAM/ROM required	Built-in RAM and flash
Power consumption	Higher	Lower
Cost	Higher	Lower
Size	Larger Circuit	Compact Circuit
Speed	very High	Moderate to High
Application	PC, Laptop, server	Robotics, IoT, automation
example	Intel Core i7	ESP32, UNO, NANO

microprocessor used for High speed computing and microcontroller used embedded control.

Ans to the Q: NO: 2

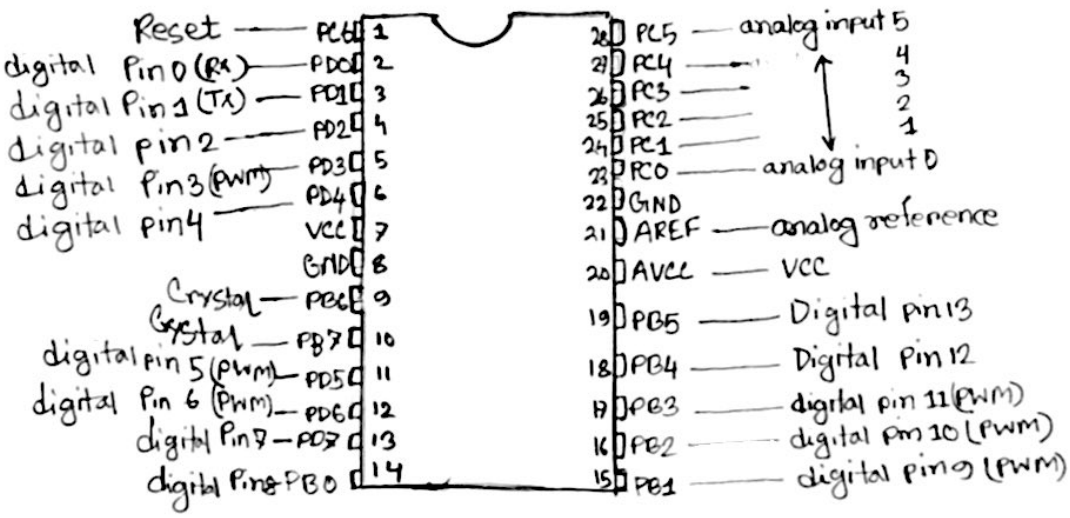

fig: Arduino UNO pin diagram.
ATMEGA 328

NOTE: Arduino Mega refers to Arduino MEGA 2560. It's Different from UNO. Mega uses ATMega 2560 and has 54 digital I/O pins 16 analog input pins, 15 PWM pins, 4 URAT ports and 256KB flash memory.

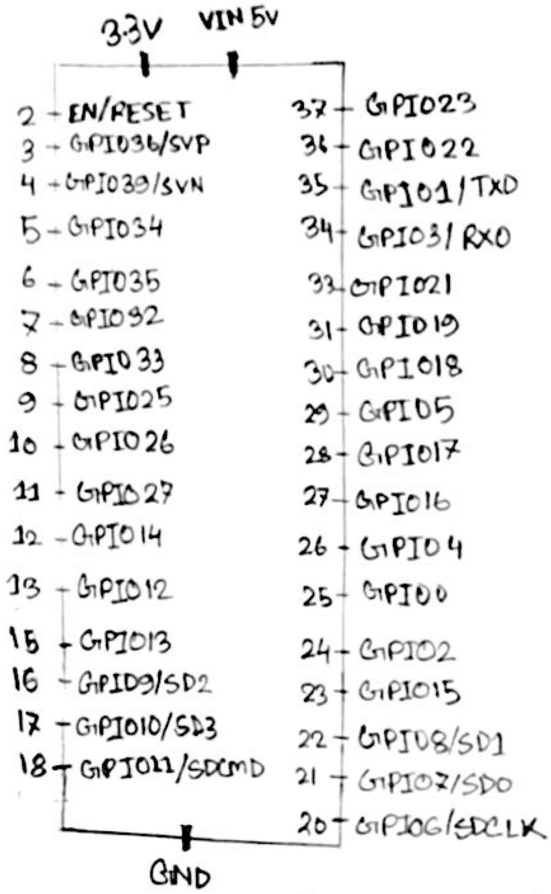

Fig: ESP32 pin Diagram

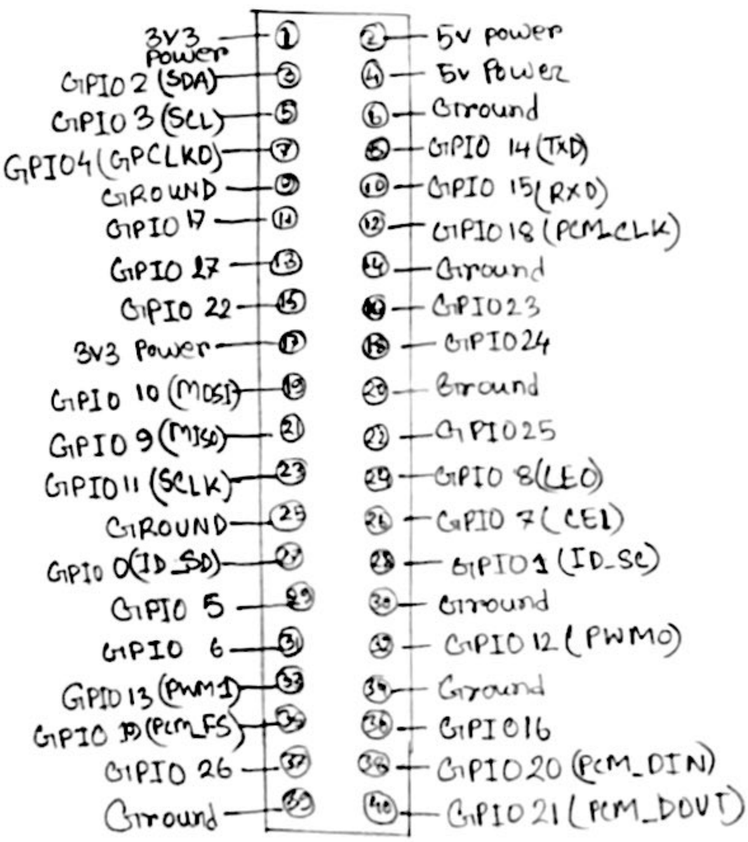

Fig: Raspberry pi 5 pin Diagram

Ans to the Q: NO: 3

for my project Iam Using ESP32 with wifi and Bluetooth Board.
My project is an AI and IoT Based Smart farming Rover. ESP32 is best Choice for the built in Wifi, and bluetooth which are very important for IOT monitoring and remote control. it can collect data from sensors such as soil moisture sensor, temperature and humidity. Sensor and send data to a mobile app or Cloud platform.
ESP32 has a 32-bit dual core processor with speed upto 240MHz So it is faster than Arduino UNO. It also has many GPIO pins, ADC pins and PWM channels. cause PWM support, it can control DC motors through a motor controller such as L298N. it can also Control Servo motors, pumps and otheractuors.
So, overall ESP32 is low price and easily available in Bangladesh also suitable for battery powerd project.

Technical Reason for choosing ESP32:-

Soil, environment monitoring → Support Analog and Digital Sensor
Motor Control → PWM pins can Control motor driver
IoT communication → Built in wifi
Remote Control → Wifi & Bluetooth Support.
Multiple Components → Enough GPIO Pins
Battery operation → Low power Consumption.
future AI upgrade → Faster than Arduino UNO.